Firewall Management System Requirements

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks. The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts. The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings. The solution should be secure, scalable, reliable, and manageable.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Policy	A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Firewall	A control that permits or restricts network traffic according to rules; mentioning it does not authorize automatic changes.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures and the SDLC plan are available in the supporting report.

2 Overall description

2.1 Product perspective

Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- Access permissions — FR-002
- Policy controls — FR-001, FR-003
- Monitoring and alerts — FR-002
- Dashboard — FR-003
- Audit logging — FR-003
- Network isolation — FR-002

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
User	guest	Uses permitted functions; each source class keeps its specified permissions.
Admin	network administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

- The solution should be secure, scalable, reliable, and manageable.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- Definition of scalability thresholds in terms of concurrent users or network traffic volume
- Quantitative reliability metrics (e.g., uptime percentage) to be met during peak operations
- Specific manageability criteria such as interface usability, configuration speed, and error recovery
- Which specific compliance framework or policy is intended to be applied to the firewall management system?
- What are the defined thresholds for scalability and reliability in terms of concurrent users, network traffic volume, and uptime percentage?
- How will manageability criteria such as interface usability, configuration speed, and error recovery be measured and validated?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: FR-001, FR-003. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.4 Communications interfaces

Related requirements: FR-001, FR-002, FR-003. Interface details beyond those requirements remain unspecified.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall provide a centralized firewall management interface to manage firewall policies across academic, administrative, research, hostel, and guest networks.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts.

Traceability: E-002, TEST-002, TEST-003, TEST-004, TEST-005

FR-003 · Should · explicit

The platform shall maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings.

Traceability: E-003, TEST-006, TEST-007, TEST-008

3.3 Performance requirements

NFR-001 · Should · explicit

The solution shall be secure, scalable, reliable, and manageable.

Traceability: E-004, TEST-009

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

Also covered by: NFR-001.

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-002, FR-003, NFR-001.

3.6.4 Maintainability

Also covered by: NFR-001.

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

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